Claudio Aracena, PhD

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Education

- **University of Chile** — **Santiago, Chile**
  PhD in Engineering Systems — *2021 – 2024*
- **University of Sydney** — **Sydney, Australia**
  Master of Information Technology and Master of Information Technology Management — *2015 – 2017*
- **University of Chile** — **Santiago, Chile**
  Industrial Engineering with Major in Information Technology — *2008 – 2013*
  Graduated with highest distinction.

Academic Appointments

- **IPAI Foundation** — **Heilbronn, Germany**
  Science Resident — *August 2026 – Present*
  - Lead the ClinPudu project: applied research and deployment of privacy-preserving small language models that run locally in healthcare institutions.
- **Business School, Adolfo Ibáñez University** — **Santiago, Chile**
  Assistant Professor, Operations and Analytics Area — *September 2025 – Present*
  - Teach AI, analytics and programming courses at undergraduate and postgraduate levels. Academic Director of the Diploma in Business and AI.
  - Associate Researcher at GobLab UAI. Academic Director of the Diploma in AI for Public Policy.
- **Faculty of Dentistry, San Sebastián University** — **Santiago, Chile**
  Lecturer–Researcher — *March 2025 – June 2025*
  - Lecturer in technology courses and member of the Data and Artificial Intelligence in Health Group.

Research Grants and Funding

- **IPAI Science Residency** – privacy-preserving clinical language models (ClinPudu) — *2026 – 2027*
  €100k+ in grant funding, as project lead.
- **Adolfo Ibáñez University research funding** — *2025 – Present*
  More than US$6k for individual research and US$8k for collaborative research.
  Collaboration with the School of Government and the Faculty of Engineering and Science. Mentor of 3 Master's students.

Publications

Journal papers

Miranda, L., Vakili, T., **Aracena, C.**, Villena, F., Dunstan, J. (2026). Conditional synthesis of training data for clinical NLP tasks in Spanish and English. *BMC Medical Informatics and Decision Making*. Accepted.

Retamales, J., **Aracena, C.**, Aguilera Beovides, L., Cook, C., Lavin, A., Lira, F., Marquez, J., Olivares-Hurtado, L., Tobar Bastías, M. J., Uribe, V., Vildoso, F., Zlatar, Z., Iturrieta-Guaita, N. (2026). Improving patient-reported end-of-life data capture: A pilot mixed-methods evaluation of an electronic patient reported outcomes chatbot platform using messaging applications from bereaved caregivers in Chile. *JCO Global Oncology*, 12(8), e2500647.

Dunstan, J., Vakili, T., Miranda, L., Villena, F., **Aracena, C.**, Quiroga, T., Vera, P., Viteri, S., and Rocco, V. A pseudonymized corpus of occupational health narratives for clinical entity recognition in Spanish (2024). *BMC Medical Informatics and Decision Making*, 24(1), 204.

Aracena, C., Villena, F., Arias, F., and Dunstan, J. Aplicaciones de aprendizaje automático en salud [Machine learning applications in health] (2022). *Revista Médica Clínica Las Condes*, 33(6), 568–575.

Workshop papers

Aracena, C., Miranda, L., Vakili, T., Villena, F., Quiroga, T., Núñez-Torres, F., Rocco, V., and Dunstan, J. A Privacy-Preserving Corpus for Occupational Health in Spanish: Evaluation for NER and Classification Tasks (2024). *6th Clinical Natural Language Processing Workshop at NAACL*.

Aracena, C., Rodríguez, N., Rocco, V., and Dunstan, J. Pre-trained language models in Spanish for health insurance coverage (2023). *5th Clinical Natural Language Processing Workshop at ACL*.

Aracena, C., and Dunstan, J. Development of pre-trained language models for clinical NLP in Spanish (2023). *Student Research Workshop at EACL*.

Aracena, C., Villena, F., Rojas, M., and Dunstan, J. A Knowledge-Graph-Based Intrinsic Test for Benchmarking Medical Concept Embeddings and Pretrained Language Models (2022). *International Workshop on Health Text Mining and Information Analysis (LOUHI) at EMNLP*.

Aracena, C., Loyola, P., Slanzi, G., Matsuo, Y., and Velásquez, J.D. Towards a Unified Replication Framework for EEG-based Emotion Classification (2015). *BrainKDD: 2nd International Workshop on Data Mining for Brain Science at ACM SIGKDD*.

Conference papers

Garcia-Flores, R., Banerjee, S., Mathews, G., Vacher, B., Thorne, B., Borhan, N., **Aracena, C.**, and Tyshetskiy, Y. Analysis of Demand and Operations of Inter-modal Terminals (2017). *National Conference of the Australian Society for Operations Research*.

Oweis, N.E., **Aracena, C.**, George, W., Oweis, M., Soori, H., and Snášel, V. Internet of Things: Overview, Sources, Applications and Challenges (2016). *2nd International Afro-European Conference for Industrial Advancement (AECIA)*.

Aracena, C., Basterrech, S., Snášel, V., and Velásquez, J.D. Neural Networks for Emotion Recognition Based on Eye Tracking Data (2015). *IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC)*.

Slanzi, G., **Aracena, C.**, and Velásquez, J.D. Eye tracking and EEG features for salient Web object identification (2015). *International Conference on Brain Informatics & Health (BIH)*.

Policy publications

Denis, G., Hermosilla, M., **Aracena, C.**, Sánchez, R., González, N., and Pombo, C. Responsible use of AI for public policy: Project formulation manual (2021). *Inter-American Development Bank*.

Teaching Experience

- **Adolfo Ibáñez University, Business School** — **Santiago, Chile**
  Assistant Professor; Academic Director, Diploma in Business and AI — *2025 – Present*
  - Undergraduate: Programming in Python, Business Analytics, Computational Finance, Operations Management.
  - Executive education: AI project formulation module of the Diploma in Business and AI.
- **Adolfo Ibáñez University, School of Government** — **Santiago, Chile**
  Academic Director, Diploma in AI for Public Policy — *2023 – Present*
  Lecturer — *2020 – Present*
  - Designed the curriculum of the diploma, including responsible AI and ethics content.
  - Machine Learning, Text Mining and Large Language Models modules, and AI and Data Science Project Scoping course.
- **University of Development, Faculty of Engineering** — **Santiago, Chile**
  Lecturer, Master of Data Science — *2018 – 2022*
  - Data Storage and Capture course.
- **Intersect Australia** — **Sydney, Australia**
  Trainer — *February 2018 – August 2018*
  - Delivered research-computing workshops using live-coding methods (Python, R, SQL, MATLAB, Bash, Git).
- **University of Chile, Department of Industrial Engineering** — **Santiago, Chile**
  Lecturer (2014); Teaching Assistant (2012 – 2013) — *2012 – 2014*
  - Lecturer: Data Warehousing. Teaching Assistant: Information Technologies, Web Mining, Introduction to Knowledge Management, Web and Business.

Teaching Development

- **Adolfo Ibáñez University** — **Santiago, Chile**
  Diploma in Learning and Teaching in Higher Education (in progress) — *2026*
- **Digital Education Council** — **Online**
  Certificate in AI for Higher Education - AI for Faculty — *2026*

Industry and Applied Research Experience

- **Chilean Safety Association (ACHS)** — **Santiago, Chile**
  Data Science Consultant — *October 2022 – January 2025*
  - Pre-training of clinical language models in Spanish and their applications to occupational health and health insurance coverage.
- **Data Science for Social Good Fellowship** — **Kaiserslautern, Germany**
  Fellow — *June – September 2023*
  - Reduced Red Cross ambulance response times through early relocation modelling.
- **GobLab UAI, Adolfo Ibáñez University** — **Santiago, Chile**
  Data Scientist — *October 2019 – December 2021*
  - Led technical design and execution of public-sector AI initiatives, including NLP analysis of citizen input for legislative processes and environmental predictive modelling.
- **Center for Mathematical Modeling, University of Chile** — **Santiago, Chile**
  Project Engineer — *October 2018 – September 2019*
  - Machine learning prediction of medical appointment no-shows at Luis Calvo Mackenna Children's Hospital.
- **Data61, CSIRO** — **Sydney, Australia**
  Associate Software Developer — *October 2015 – December 2016*
  - Software development for the 64knots team (freight and shipping optimisation and machine learning).
- **Earlier research and industry roles** — *2013 – 2015*
  - Research Intern, VSB–Technical University of Ostrava, Czech Republic (2014–2015): neural networks for emotion recognition from pupil dilation.
  - Project Engineer, Web Intelligence Centre, University of Chile (2014): EEG, eye tracking and machine learning.
  - Intern, Infosys Limited, Bangalore, India (2013): tweet classification and database design.

Technical Skills

Languages: Python, R, SQL, JavaScript, LaTeX.
AI and data: HuggingFace Transformers, scikit-learn.
Infrastructure: Git, Docker, Linux.

Distinctions

- **ANID Scholarship for National PhD Programs**, Chile (2021–2024).
- **Becas Chile Scholarship**, Conicyt (2015–2017): awarded for postgraduate study at a leading foreign university based on academic excellence.
- **University of Chile Scholarship** (2007–2013): awarded to the top 100 admitted students.

Languages

Spanish (native) · English (professional proficiency)